

1 Num and 1 Cat Var Guided Lab 2025

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Exploring the Data

```
data(Births2015)
head(Births2015)
```

```
##           date births wday year month day_of_year day_of_month day_of_week
## 1 2015-01-01   8068  Thu 2015     1           1           1           5
## 2 2015-01-02  10850  Fri 2015     1           2           2           6
## 3 2015-01-03   8328  Sat 2015     1           3           3           7
## 4 2015-01-04   7065  Sun 2015     1           4           4           1
## 5 2015-01-05  11892  Mon 2015     1           5           5           2
## 6 2015-01-06  12425  Tue 2015     1           6           6           3
```

I predict that it is unlikely for a specific day to have the greatest amount of child births due to the randomness of labor.

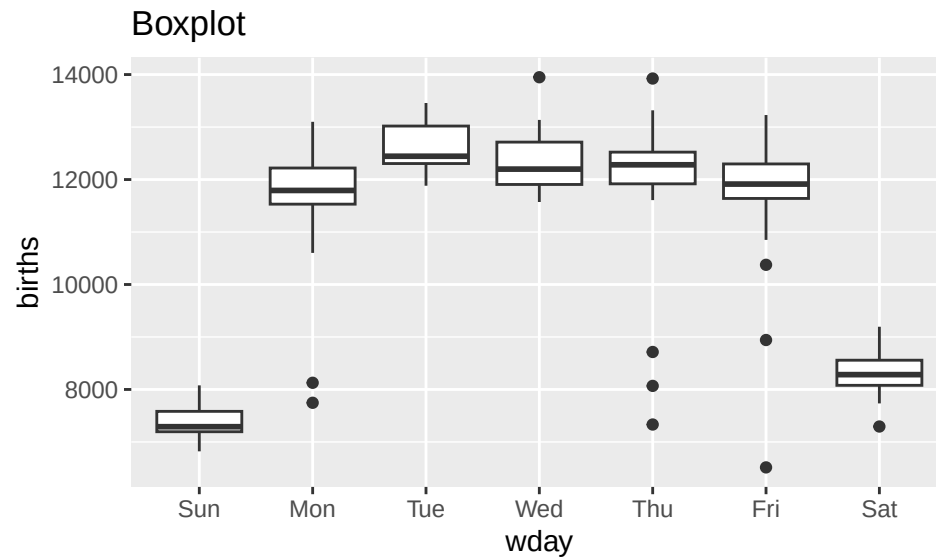
Y and X Graphs

```
require(ggforce)
```

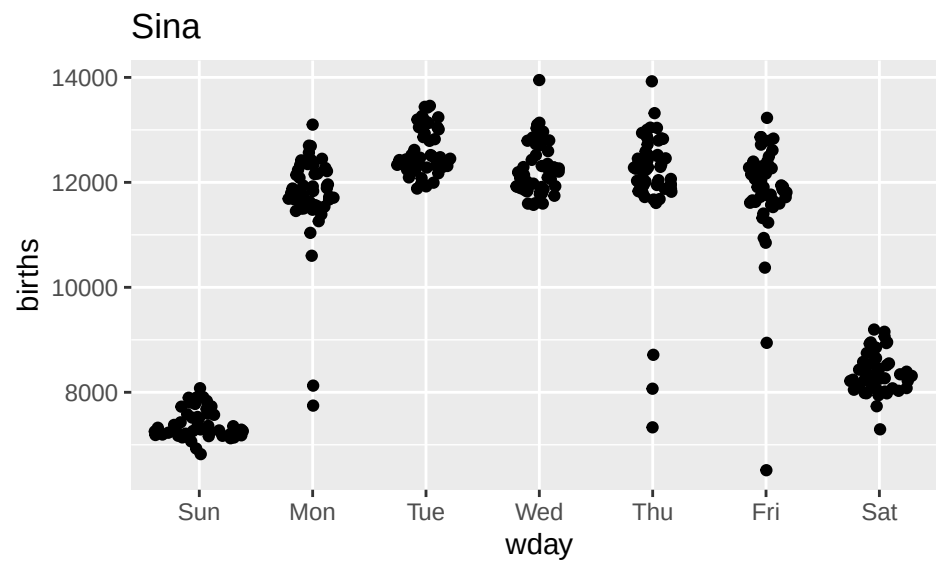
```
## Loading required package: ggforce
```

```
## Warning: package 'ggforce' was built under R version 4.4.3
```

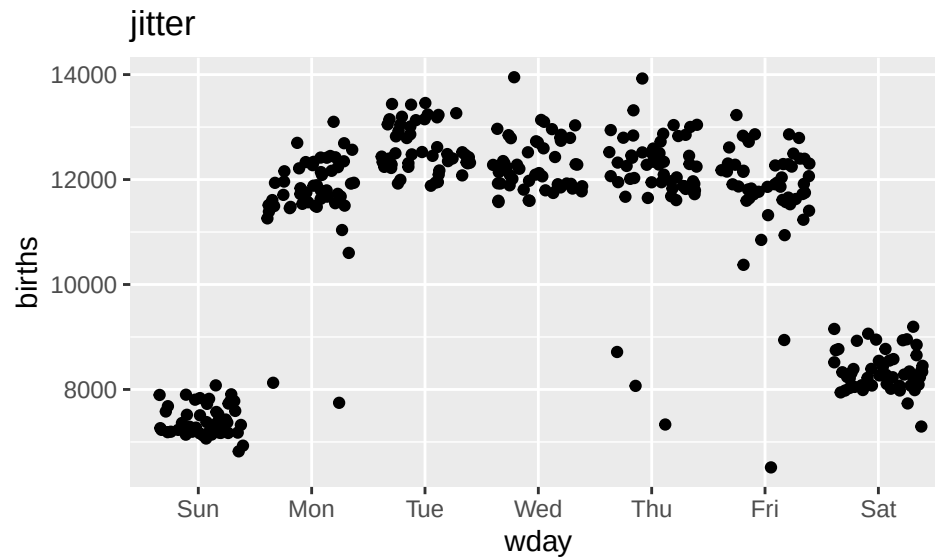
```
gf_boxplot( births ~ wday , data = Births2015 , options,title="Boxplot")
```



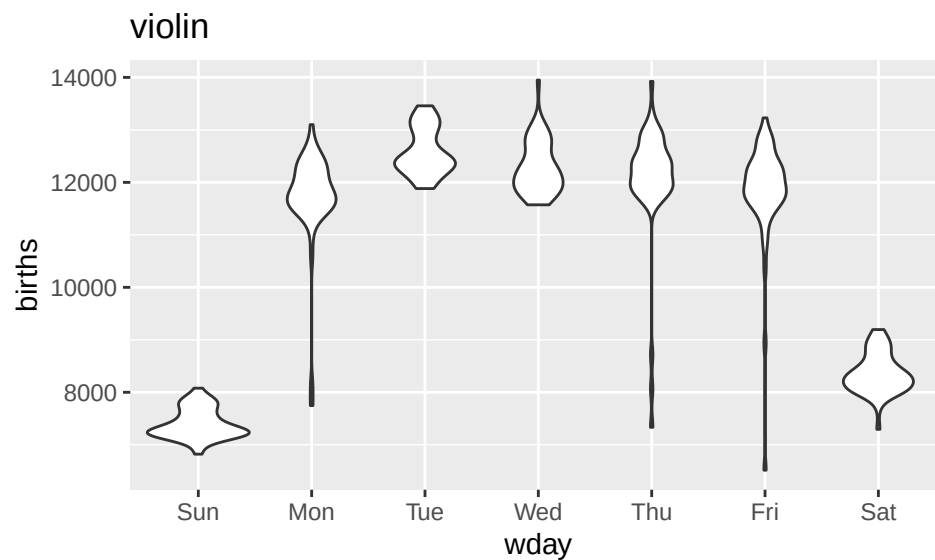
```
gf_sina( births ~ wday , data = Births2015,title="Sina")
```



```
gf_jitter( births ~ wday , data = Births2015 ,title="jitter")
```



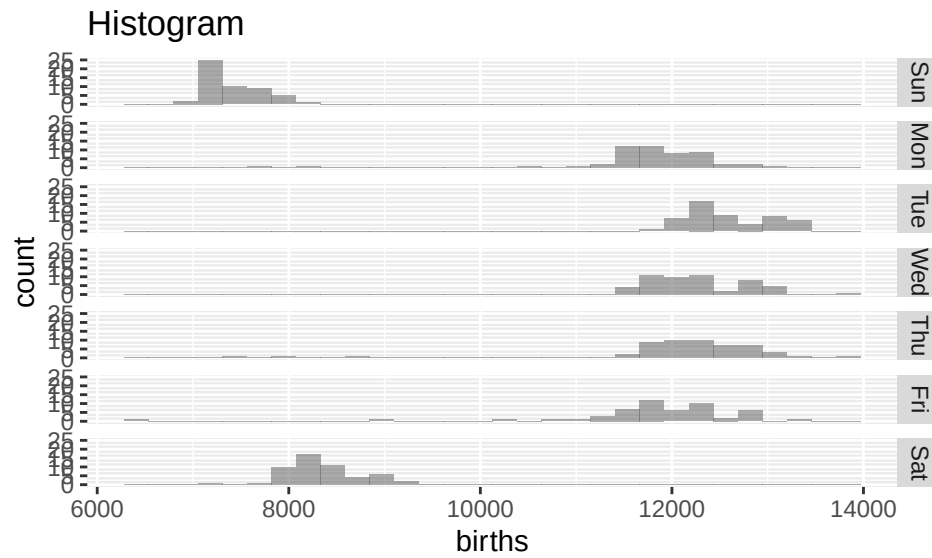
```
gf_violin( births ~ wday , data = Births2015 ,title="violin")
```



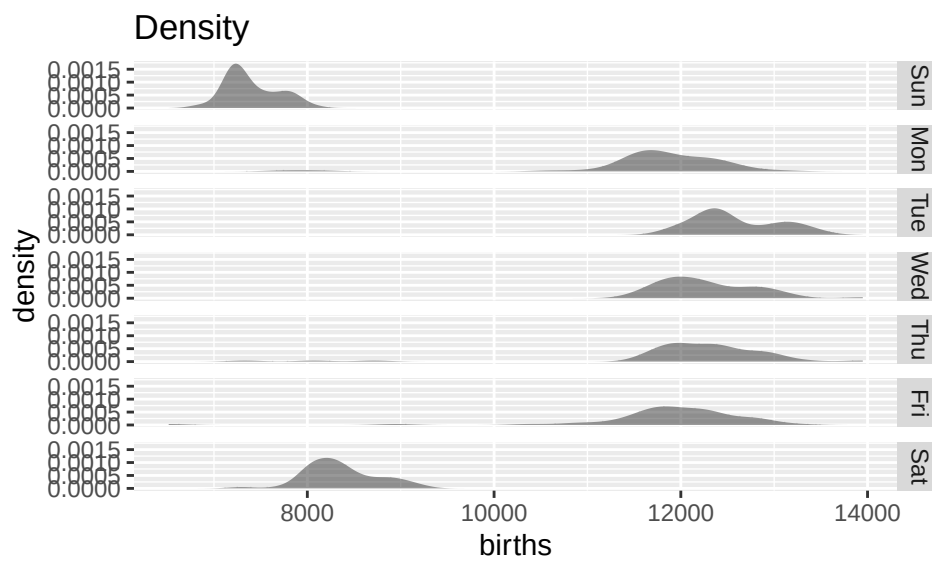
vs X graphs

```
gf_histogram(~births, data = Births2015,title="Histogram") %>%
gf_facet_wrap(~wday, ncol=1, strip.position = "right")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

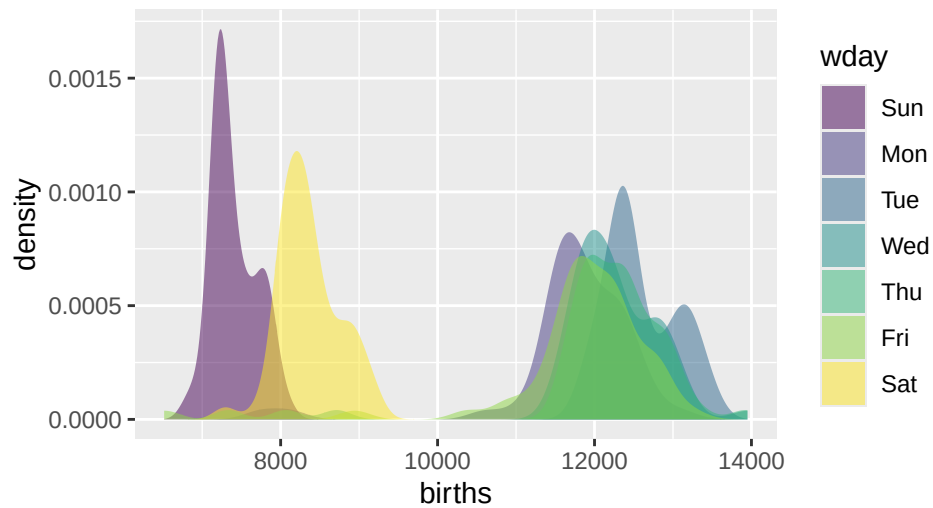


```
gf_density(~births, data = Births2015, title="Density") %>%
gf_facet_wrap(~wday, ncol=1, strip.position = "right")
```



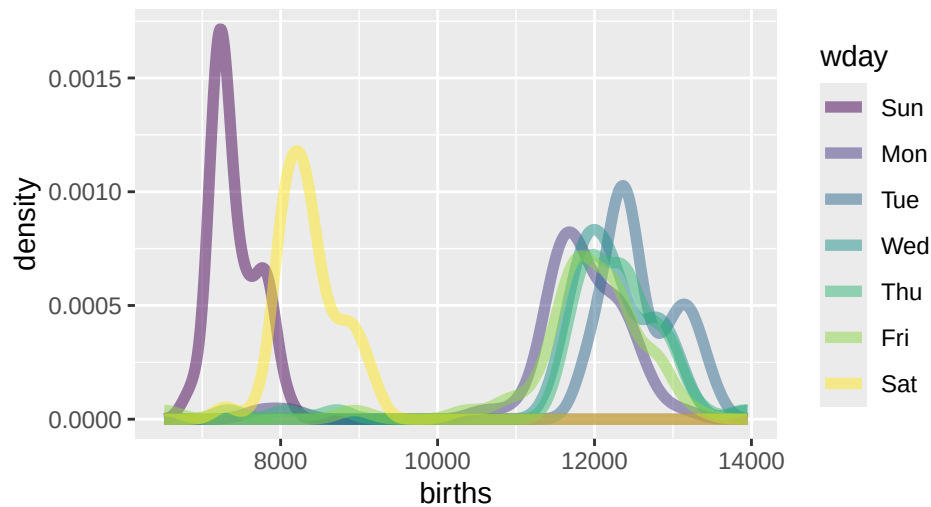
```
gf_density( ~ births , data = Births2015 , fill = ~ wday , alpha = 0.5, title="NOT faceted, "filled")
```

NOT faceted, "filled"



```
gf_dens( ~ births , data = Births2015 , color = ~ wday , size = 2, title= "colored by day of the week")
```

colored by day of the week



```
library(ggribes)
```

```
## Warning: package 'ggribes' was built under R version 4.4.3
```

```
gf_density_ridges(wday~births, data=Births2015, fill=~wday, show.legend = FALSE, scale = 2, alpha = 0.5)
```

```
## Picking joint bandwidth of 177
```

